

Hamza Idrees

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Profile

- Backend Developer with 1.5+ years of production experience building SaaS APIs serving **400+ businesses across 1,000+ locations** in 10+ countries.
- Specialized in multi-tenant architectures, AI integrations (OpenAI, Anthropic Claude), and scalable background job systems handling thousands of daily operations.
- Experienced across the full backend lifecycle — database design, API development, cloud infrastructure (AWS Lambda, S3, SQS), and automated testing with 90%+ code coverage.

Skills

Backend

JavaScript, Node.js, Express.js, Python, FastAPI, BullMQ, node-cron

Cloud

AWS Lambda, S3, SQS

Testing

Jest, Mocha, Supertest

Databases

PostgreSQL, MongoDB, Redis, Qdrant (vector search)

Integrations

OpenAI, Anthropic Claude, DeepL, SendGrid, Google APIs, RAG pipelines

Frontend

React, TypeScript, Tailwind CSS, Redux Toolkit

Professional Experience

Backend Developer

Global Software Consulting

2024/12 – Present

Project: Obenan

SaaS platform for business location management and online reputation across Google, Facebook, and 20+ review platforms.

API DEVELOPMENT & DATABASE MANAGEMENT

- Built and maintained RESTful API endpoints using Node.js and Express.js, documented with Swagger/OpenAPI.
- Developed features across a dual-database architecture: PostgreSQL (Sequelize ORM, 100+ models) and MongoDB (Mongoose) serving a multi-tenant platform.
- Added Redis caching layers to high-traffic endpoints, eliminating redundant database queries on repeated requests.

AI & THIRD-PARTY INTEGRATIONS

- Developed AI-powered sentiment analysis and review response pipeline using OpenAI and Anthropic Claude APIs, automating analysis for businesses across multiple countries.
- Built a batched OpenAI post description generation pipeline processing **100+ posts in under 30 seconds**, eliminating API timeout issues and reducing generation time by over 90%.
- Integrated DeepL for multi-language translation and Google Business Profile sync for reviews, insights, and posts across multiple locations.
- Built transactional email system using SendGrid and AWS SES with webhook event tracking and delivery monitoring.

BACKGROUND PROCESSING & SCHEDULING

- Developed and maintained 10+ BullMQ job queues with Redis for async processing — handling review ingestion, sentiment analysis, report exports, and email delivery.
- Automated scheduled jobs for report scheduling, Datashake review dumps, sentiment analysis batching, and two-way location listing sync with Google/Facebook.

AUTHENTICATION & SECURITY

- Developed JWT-based authentication and role-based access control for multi-tenant user management.
- Integrated OAuth 2.0 flows (Google, Facebook) and led the integration of Microsoft Entra ID (Azure AD) for enterprise SSO.
- Enforced security standards including input validation, rate limiting, and webhook signature verification.

CLOUD INFRASTRUCTURE & TESTING

- Engineered a Lambda function to fetch **40,000+ reviews within AWS's 15-minute execution limit** using optimized batching and pagination — overcoming one of Lambda's hardest constraints at scale.
- Managed file storage with AWS S3 using presigned URLs and async messaging with AWS SQS.
- Wrote unit and integration tests using Jest, Mocha, and Supertest, contributing to 90%+ code coverage.

FRONTEND DEVELOPMENT

- Owned the Report Schedule Records module end-to-end (React, TypeScript, Tailwind CSS, NextUI, Vite) — delivering filtering, debounced search, sortable pagination, and full CRUD operations.

Tech Stack: Node.js, Express.js, PostgreSQL (Sequelize), MongoDB (Mongoose), Redis, BullMQ, AWS Lambda/S3/SQS, JWT, OAuth 2.0, React, TypeScript, Tailwind CSS

Project: Lumen

Production-grade RAG system enabling clinic staff to query ~5,000 patient records in plain English and receive grounded, real-time cited answers.

RAG PIPELINE & AI INTEGRATION

- Designed and implemented an end-to-end RAG pipeline using Python and FastAPI: GPT-4o-mini for structured intent extraction, text-embedding-ada-002 for vector embeddings, and GPT-4.1-mini for streaming LLM responses over SSE — delivering first-token responses in under 2 seconds.
- Built a 20+ filter query parser extracting condition, age range, doctor, admission/discharge dates, billing range, blood type, and more from free-text queries — reducing irrelevant retrievals by over 60% compared to unfiltered semantic search via pre-filtered Qdrant lookups.
- Integrated Qdrant as a vector store ingesting 5,000 patient records as natural-language embeddings with rich metadata, achieving sub-100ms vector search response times for hybrid semantic + metadata retrieval.

BACKEND & API DEVELOPMENT

- Developed RESTful FastAPI endpoints for chat (SSE streaming), CSV ingestion, health checks, stats, and PDF report export — supporting bulk ingestion of 5,000 records in under 60 seconds with admin token authentication.
- Implemented cosine similarity thresholding and a "don't know" fallback, eliminating fabricated answers on out-of-scope queries and reducing hallucination to near-zero on insufficient context.

FRONTEND DEVELOPMENT

- Built a Next.js 14 App Router frontend with a real-time streaming chat interface, citation cards linking each answer to exact patient rows, and a PDF export feature using Tailwind CSS and Framer Motion.

Tech Stack: Python, FastAPI, Qdrant, OpenAI (GPT-4o-mini, text-embedding-ada-002, GPT-4.1-mini), SSE, Next.js 14, Tailwind CSS, Framer Motion

Projects

ShipEase | React, Node.js, MUI, PostgreSQL

Logistics Management Platform

- Built a full-stack logistics platform connecting service providers, drivers, and users with real-time shipment tracking and seamless communication.
- Developed the web app, backend API, and mobile application for both user and driver actors.

Education

2020/09 – 2024/06

Bachelor in Software Engineering

FAST - National University of Computer and Emerging Sciences